

1. Project Overview

This document outlines the deployment, configuration, troubleshooting, and validation of a **Windows Server 2019 Active Directory lab environment**.

The goal of the lab was to successfully configure a **Domain Controller** with the following core services:

- **Active Directory Domain Services (AD DS)**
- **DNS (Domain Name System)**
- **DHCP (Dynamic Host Configuration Protocol)**

The lab also includes a **Windows 11 client** used to validate:

- DHCP lease assignment
- DNS name resolution
- Domain Controller discovery
- readiness for domain join

2. Lab Environment

2.1 Server Configuration

- **Server OS:** Windows Server 2019
- **Initial Server Name:** Server_2019_DC
- **Final Server Name:** Server 2019 DC01
- **Server IP Address:** 10.10.10.1
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** [Isolated LAB]
- **Preferred DNS Server:** 10.10.10.1 (self-referencing as Domain Controller)

2.2 Domain Configuration

- **Domain Name:** local.lab
- **Forest Name:** local.lab
- **Forest Functional Level:** Windows Server 2016
- **Domain Functional Level:** Windows Server 2016

2.3 Client Configuration

- **Client OS:** Windows 11

- **Client Name:** PC_01_Local
- **IP Assignment:** DHCP
- **DNS Server Assigned via DHCP:** 10.10.10.1
- **DHCP Server:** 10.10.10.1

3. Objectives

The objectives of this lab were:

1. Deploy a Windows Server 2019 machine as a Domain Controller
2. Install and configure:
 - a. AD DS
 - b. DNS
 - c. DHCP
3. Validate that the Windows 11 client receives:
 - a. IP address
 - b. DNS settings
 - c. DHCP lease
4. Validate DNS resolution from the client to the server
5. Troubleshoot domain controller discovery issues
6. Confirm successful DC locator and Active Directory service discovery
7. Prepare the lab for:
 - a. domain join
 - b. user and OU creation
 - c. Group Policy testing
 - d. future Entra ID / Intune learning

4. DHCP Configuration and Validation

The DHCP service was configured on the Windows Server 2019 Domain Controller to provide IP configuration to the Windows 11 client.

The client successfully obtained:

- an IP address in the lab subnet
- DNS server address 10.10.10.1
- DHCP server address 10.10.10.1

This confirmed that the DHCP scope was functioning correctly and that the client could communicate with the server.

4.1 Evidence

[Windows 11 client receiving IP configuration from DHCP, including DHCP server and DNS server set to 10.10.10.1.]

```
Windows IP Configuration

Host Name . . . . . : DESKTOP-FICTFCB
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : local.lab

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : local.lab
Description . . . . . : Microsoft Hyper-V Network Adapter
Physical Address. . . . . : 00-15-5D-01-8A-03
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::bd2d:9cb0:80a3:5091%14(Preferred)
IPv4 Address. . . . . : 10.10.10.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 08 April 2026 01:55:11
Lease Expires . . . . . : 16 April 2026 01:55:11
Default Gateway . . . . . :
DHCP Server . . . . . : 10.10.10.1
DHCPv6 IAID . . . . . : 83891549
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-65-38-7A-00-15-5D-01-8A-03
DNS Servers . . . . . : 10.10.10.1
NetBIOS over Tcpip. . . . . : Enabled
```

figure 1

4.2 Notes

During initial validation, the client received an IP address successfully.

At one stage, the client showed **no default gateway**, which was noted as a lab configuration detail.

Since this is an isolated test environment, the lack of a gateway did not prevent internal AD/DNS/DHCP testing inside the lab subnet.

5. Initial Connectivity Testing

After DHCP assignment, network connectivity was tested from the Windows 11 client.

5.1 Ping Test to Domain Controller IP

The client successfully pinged the server IP address:

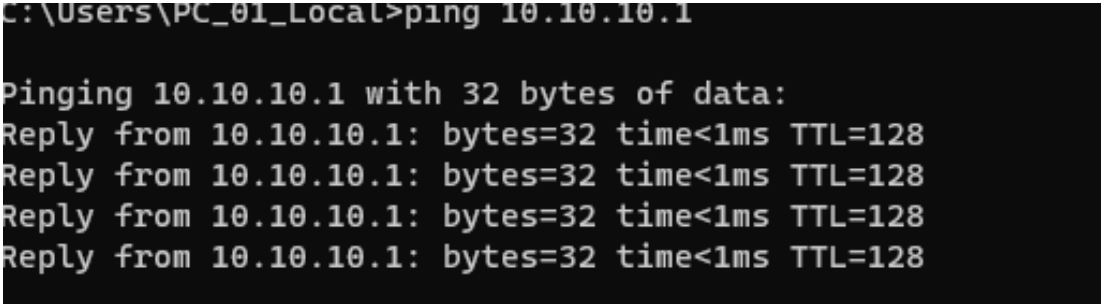
- 10.10.10.1

This confirmed:

- Layer 3 connectivity
- basic network communication
- client-to-server reachability within the lab network

5.2 Evidence

[Successful ICMP connectivity from Windows 11 client to the Domain Controller IP address (10.10.10.1).



```
C:\Users\PC_01_Local>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
Reply from 10.10.10.1: bytes=32 time<1ms TTL=128
```

figure 2

6. DNS Name Resolution Testing

After verifying IP connectivity, DNS name resolution was tested from the client.

The client was able to resolve the server hostname using the DNS service hosted on the Domain Controller.

Initial testing used the original server name:

- Server_2019_DC

Later, the server was renamed to:

- Server_2019_DC01

This rename created a temporary mismatch between the expected hostname and the DNS / AD registration state, which required troubleshooting and revalidation.

6.1 Initial DNS Validation

The client was able to query the DNS server, but the response was not healthy due to timeout-related issues during initial testing.:

- 10.10.10.1

The DNS query initially timed out; after troubleshooting the timeout issue was resolved. This confirmed that the DNS service was reachable and responding.

6.2 Evidence

DNS resolution from the Windows 11 client to the Domain Controller using the lab DNS server (10.10.10.1) not healthy.

```
C:\Users\PC_01_Local>nslookup Server_2019_dc
DNS request timed out.
    timeout was 2 seconds.
Server:    UnKnown
Address:   10.10.10.1

Name:      Server_2019_dc.local.lab
Address:   10.10.10.1
```

Figure 3

Resolved - DNS resolution from the Windows 11 client

```
C:\Users\PC_01_Local>nslookup
Default Server:  server_2019_dc.local.lab
Address: 10.10.10.1

>
>
C:\Users\PC_01_Local>
C:\Users\PC_01_Local>
C:\Users\PC_01_Local>nslookup server_2019_dc
Server:  server_2019_dc.local.lab
Address: 10.10.10.1

Name:    server_2019_dc.local.lab
Address: 10.10.10.1
```

Figure 4

7. Issue Encountered – Domain Controller Discovery Failure

Although basic DNS resolution was working, the client initially failed to locate the Domain Controller using Active Directory DC locator mechanisms.

The following command was used:

```
nltest /dsgetdc:local.lab
```

7.1 Initial Result

The command returned:

- **ERROR_NO_SUCH_DOMAIN (1355)**

This indicated that while the client could resolve the server name, it was not yet able to discover the Domain Controller via the required **AD-specific DNS SRV records**.

This is a more advanced validation than simple hostname resolution and depends on:

- proper DNS registration
- AD DS service registration
- Netlogon SRV record publication
- server rename consistency after reboot

7.2 Evidence

nltest failure / ERROR 1355

```
C:\Users\PC_01_Local>nltest /dsgetdc:local.lab  
Getting DC name failed: Status = 1355 0x54b ERROR_NO_SUCH_DOMAIN
```

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8. Root Cause Analysis

The issue was traced to a combination of the following:

1. **Server rename timing**
 - a. The server was initially named Server_2019_DC
 - b. It was later renamed to Server_2019_DC01**
 - c. Some DNS / AD records were still referencing the previous hostname until reboot and re-registration
2. **DNS record cleanup / registration**
 - a. Forward lookup and reverse lookup consistency needed to be validated

- b. Domain Controller service records needed to be re-registered
- 3. **Post-promotion registration timing**
 - a. After Domain Controller promotion, some AD-integrated DNS SRV records may require:
 - i. reboot
 - ii. Netlogon restart
 - iii. DNS re-registration

9. Troubleshooting Actions Performed

The following corrective actions were taken:

9.1 Verified DNS Forward Resolution

Confirmed that the client could resolve the server via DNS and that the DNS server responded correctly.

Example checks included:

```
nslookup
nslookup server_2019_dc
ping server_2019_dc.local.lab
```

9.2 Reviewed Forward and Reverse DNS Zones

The DNS configuration was reviewed to ensure:

- Forward Lookup Zone for `local.lab` existed
- Host (A) records pointed to `10.10.10.1`
- Reverse Lookup Zone was present and aligned with the subnet
- PTR records were correctly mapped where applicable

9.3 Rebooted the Domain Controller

A full reboot was performed after the server rename so that:

- the new hostname would fully apply
- AD DS services would reinitialize
- Netlogon could republish SRV records
- DNS-integrated records could synchronize correctly

9.4 Revalidated DNS and AD Registration

After reboot, DNS and DC locator tests were repeated.

10. Successful Post-Reboot Validation

After the reboot and DNS/AD record correction, the client was able to successfully locate the Domain Controller.

The following command was used again:

```
nltest /dsgetdc:local.lab
```

10.1 Successful Result

The command returned a successful DC discovery response showing:

- Domain Controller name
- Domain name
- Forest name
- Site name
- Key service flags including:
 - PDC
 - GC
 - LDAP
 - KDC
 - WRITABLE
 - DNS_DC
 - DNS_DOMAIN
 - DNS_FOREST

This confirmed that:

- AD DS was operational
- DNS was properly serving AD locator records
- the client could discover the Domain Controller correctly
- the environment was ready for domain join and further AD testing

10.2 Evidence

Successful Domain Controller discovery using nltest, confirming working AD DS, DNS SRV records, LDAP, Kerberos, and Global Catalog services.

successful nltest /dsgetdc:local.lab output

```
C:\Users\PC_01_Local>
C:\Users\PC_01_Local>
C:\Users\PC_01_Local>nltest /dsgetdc:local.lab
        DC: \\Server_2019_DC01.local.lab
        Address: \\10.10.10.1
        Dom Guid: 2ce4842e-f0ab-4a42-9b74-38a436191121
        Dom Name: local.lab
        Forest Name: local.lab
        Dc Site Name: Default-First-Site-Name
        Our Site Name: Default-First-Site-Name
        Flags: PDC GC DS LDAP KDC TIMESERV WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE FULL_SECRET WS_DS_8 DS_9 DS
10 KEYLIST
The command completed successfully
```

figure 6

11. Final Validation Summary

At the conclusion of troubleshooting, the following core services were confirmed as working:

11.1 DHCP

- Client successfully received an IP address from the DHCP scope
- DNS server and DHCP server were correctly assigned as 10.10.10.1

11.2 DNS

- Client successfully resolved the Domain Controller hostname
- Forward lookup and AD-related DNS registration were functional
- Domain Controller could be discovered through DNS-based locator records

11.3 Active Directory Domain Services

- The Domain Controller was successfully discoverable

- LDAP and Kerberos services were active
- Global Catalog was available
- The Domain Controller was writable and advertising correctly

12. Key Commands Used During Validation

12.1 Client-side Commands

```
ipconfig /all
ping 10.10.10.1
ping server_2019_dc.local.lab
nslookup
nslookup server_2019_dc
nltest /dsgetdc:local.lab
```

12.2 Useful Server-side Validation Commands

```
hostname
ipconfig /registerdns
nltest /dsregdns
net stop netlogon
net start netlogon
dcdiag /test:advertising
dcdiag /test:dns
```

13. Lessons Learned

This lab highlighted several important Active Directory deployment lessons:

1. **Basic DNS resolution is not the same as successful AD DC discovery**
 - a. A hostname may resolve, but DC locator can still fail if SRV records are not correctly registered
2. **Server renaming after deployment requires careful revalidation**
 - a. Rebooting after rename is critical
 - b. DNS and Netlogon registration may need to refresh
3. **AD depends heavily on DNS health**
 - a. Active Directory is tightly integrated with DNS
 - b. DNS misconfiguration can prevent domain join, authentication, and Group Policy processing
4. **nltest /dsgetdc is a strong real-world validation tool**

- a. It confirms actual Domain Controller discovery, not just simple name resolution

14. Next Planned Steps

The next steps for this lab are:

1. Join the Windows 11 client to the local.lab domain
2. Create:
 - a. Organizational Units (OUs)
 - b. test users
 - c. test groups
3. Validate domain login
4. Test Group Policy application
5. Prepare the environment for future learning around:
 - a. hybrid identity concepts
 - b. Microsoft Entra ID
 - c. Intune device management workflows
 - d. enterprise endpoint administration

15. Conclusion

This lab successfully demonstrated the deployment and validation of a **Windows Server 2019 Active Directory environment** with **DHCP, DNS, and AD DS**.

Despite an initial issue caused by hostname transition and incomplete Domain Controller registration, the problem was identified, corrected, and fully validated.

The final state of the environment confirms that:

- the Windows 11 client receives network settings from DHCP
- DNS resolves the Domain Controller correctly
- the client can locate the Domain Controller using AD-specific discovery mechanisms
- the lab is ready for domain join, user management, GPO testing, and future hybrid identity experimentation

This project serves as a practical foundation for demonstrating skills in:

- Windows Server administration
- Active Directory

- DNS troubleshooting
- DHCP configuration
- client/server validation
- enterprise lab documentation

16. Skills Demonstrated

- Windows Server 2019 administration
- Active Directory Domain Services deployment
- DNS forward/reverse lookup troubleshooting
- DHCP scope validation
- Domain Controller discovery troubleshooting
- Client connectivity validation
- Command-line troubleshooting (nslookup, ping, nltest, dcdiag)
- Lab documentation and root cause analysis